

UNLOCKABLE

SMART DOOR LOCK

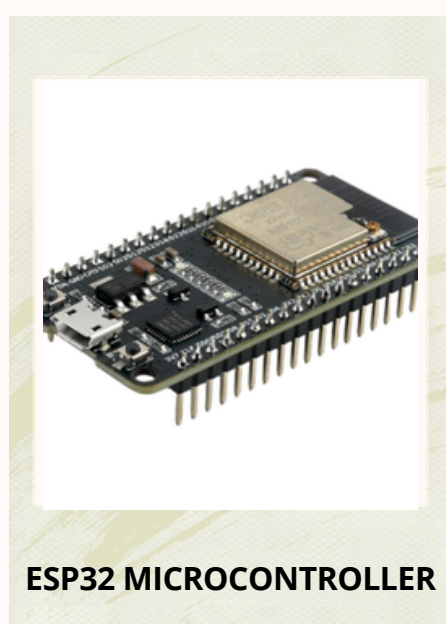
SYSTEM OVERVIEW

Unlockable is a secure door lock system that uses a PIN-based authentication method to control access. Users enter a password through push buttons, and the system verifies it before unlocking. It includes attempt tracking, lockout protection, a feature to change passwords.

COMPONENTS



SERVO MOTOR



ESP32 MICROCONTROLLER



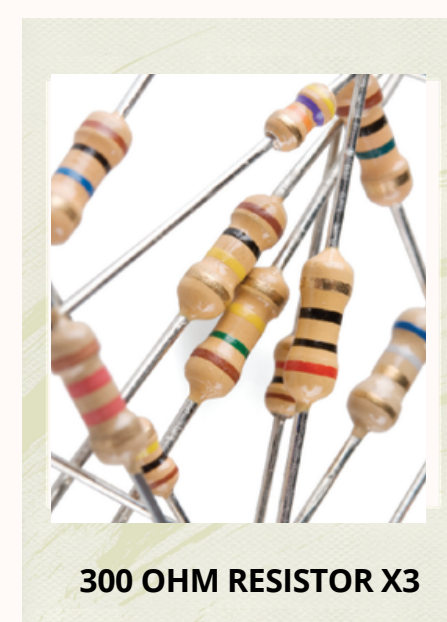
LCD DISPLAY



PUSHBUTTONS X6



PIEZO BUZZER



300 OHM RESISTOR X3



BUCK CONVERTER



RED LEDS X3

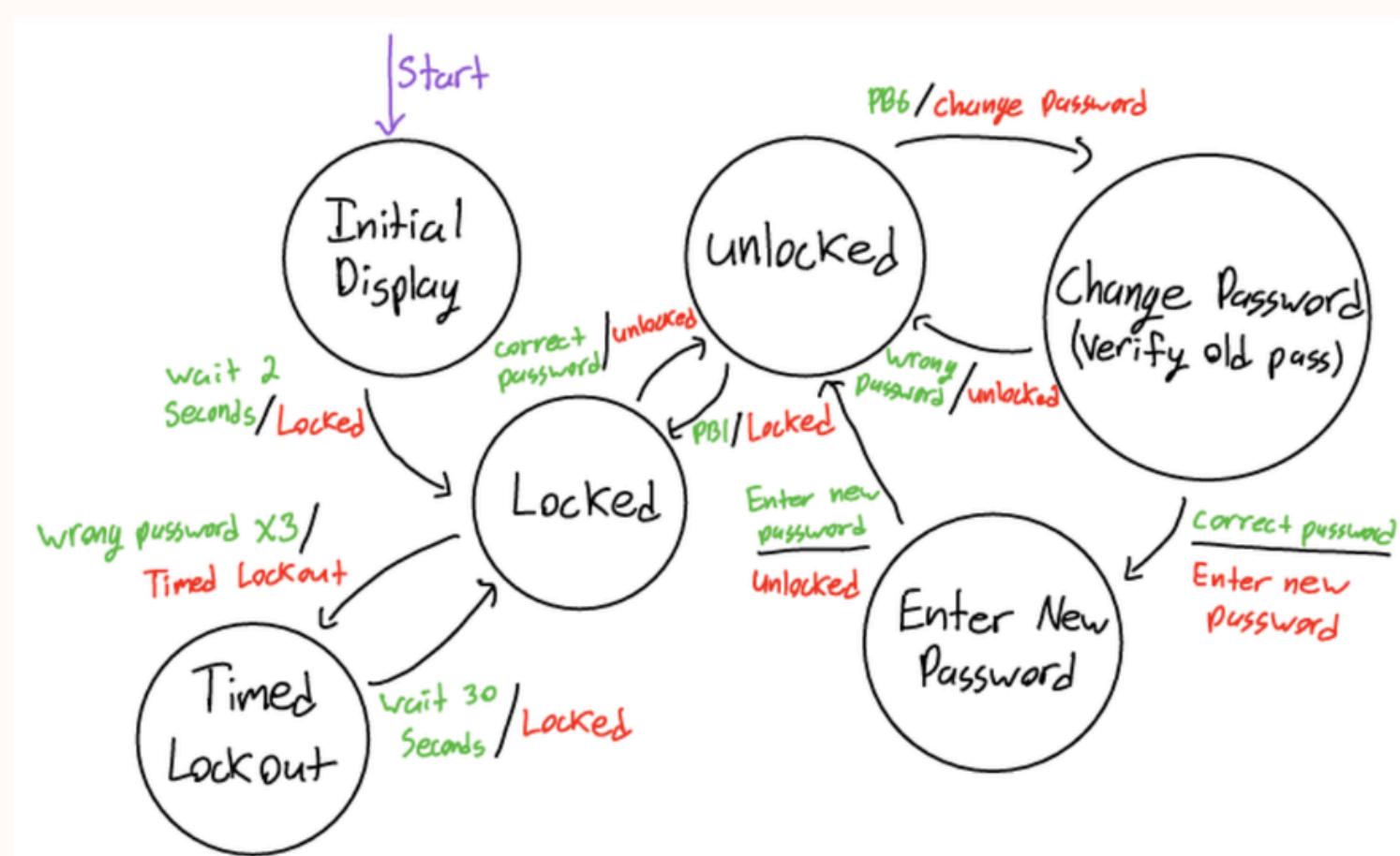


9V BATTERY

HOW IT WORKS

- System powers on in Locked state
- User enters 4-digit PIN using push buttons
- Each digit is stored and displayed with indicators on LCD display
- After 4 digits, system verifies input
- If correct, servo unlocks and access is granted
- If incorrect, buzzer sounds and attempt count increases
- After 3 failed attempts, 60-second lockout activates
- System can be manually locked or PIN changed when unlocked

FSM DIAGRAM



VIDEO DEMO

